52. The positive integers less than p^r and relatively prime to p^r are those that are not multiples of p. There are $p^{r-1}-1$ multiples of p less than p^r. Thus we see that there are $(p^r-1)-(p^{r-1}-1) = p^r-p^{r-1} = p^{r-1}(p-1)$ positive integers less than p^r and relatively prime to p^r.

53. It is no loss of generality to supppose that $G=\mathbb{Z}_n$ and that we are considering the equation $mx=0$ for a positive integer m dividing n. Clearly $0, n/m, 2n/m, \cdots, (m-1)n/m$ are m solutions of $mx=0$. If r is any solution in $\mathbb{Z}_n$ of $mx=0$, then n is a divisor of mr, so that $mr=qn$. But then $r=q(n/m)<n$, so that q must be one of $0,1,2,\cdots,m-1$, and we see that the solutions exhibited above are indeed all the solutions.

54. There are exactly d solutions, where d is the gcd of m and n. Working in $\mathbb{Z}_n$ again, we see that $0, n/d, 2n/d, \cdots, (d-1)n/d$ are solutions of $mx=0$. If r is any solution, then n divides mr so that $mr=nq$ and $r=nq/m$. Write $m=m_1d$ and $n=n_1d$ so that the gcd of m_1 and n_1 is 1. Then $r=nq/m$ can be written as $r=n_1dq/m_1d=n_1q/m_1$. Since m_1 and n_1 are relatively prime, we conclude that m_1 divides q; let $q=m_1s$. Then $r=n_1q/m_1=n_1m_1s/m_1=n_1s=(n/d)s$. Since $r<n$, we have $n_1s<n=n_1d$ so $s<d$. consequently, s must be one of the numbers $0,1,2,\cdots,d-1$ and we see that the solutions exhibited above are indeed all the solutions.

55. All positive integers less than p are relatively prime to p because p is prime, and hence they all generate $\mathbb{Z}_p$. Thus $\mathbb{Z}_p$ has no proper cyclic subgroups, and thus no proper subgroups, because as a cyclic group, $\mathbb{Z}_p$ has only cyclic subgroups.

56. **a.** Let a be a generator of H and let b be a generator of K. Because G is abelian, we have $(ab)^{rs} = (a^r)^s(b^s)^r = e^re^s = e$. We claim that no lower power of ab is equal to e, for suppose that $(ab)^n = a^nb^n = e$. Then $a^n = b^{-n} = c$ must be an element of both H and K, and thus generates a subgroup of H of order dividing r which must also be a subgroup of K of order dividing s. Because r and s are relatively prime, we see that we must have $c=e$, so $a^n=b^n=e$. But then n is divisible by both r and s, and because r and s are relatively prime, we have $n=rs$. Thus ab generates the desired cyclic subgroup of G of order rs.

b. Let d be the gcd of r and s, and let $s=dq$ so that q and r are relatively prime and $rq=rs/d$ is the least common multiple of r and s (see Exercise 47c). Let a and b be generators of H and K respectively. Then $|\langle a\rangle| = r$ and $|\langle b^d\rangle| = q$ where r and q are relatively prime. Part(**a**) shows that the element ab^d generates a cyclic subgroup of order rq which is the least common multiple of r and s.

7. Generators and Cayley Digraphs

1. 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 **2.** 0, 2, 4, 6, 8, 10 **3.** 0, 2, 4, 6, 8, 10, 12, 14, 16

4. 0, 6, 12, 18, 24, 30 **5.** $\cdots, -24, -18, -12, -6, 0, 6, 12, 18, 24, \cdots$

6. $\cdots, -15, -12, -9, -6, -3, 0, 3, 6, 9, 12, 15, \cdots$

7. **a:** Starting at the vertex a^2b, we travel three solid lines in the direction of the arrow, arriving at a^3b.

b. Starting at the vertex ab, we travel three solid lines in the direction of the arrow and then one dashed line, arriving at a^2.

c. Starting at the vertex b, we travel two solid lines in the direction of the arrow and then one dashed line, arriving at a^2.

8.

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

9. (See the answer in the text.)

10.

	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	c	f	e	b	d
b	b	d	e	f	a	c
c	c	e	d	a	f	b
d	d	f	c	b	e	a
f	f	b	a	d	c	e

11. Choose a pair of generating directed arcs, call them *arc1* and *arc2*. Start at any vertex of the digraph, and see if the sequences *arc1, arc2* and *arc2, arc1* lead to the same vertex. (This corresponds to asking if the two corresponding group generators commute.) The group is commutative if and only if these two sequences lead to the same vertex for every pair of generating directed arcs.

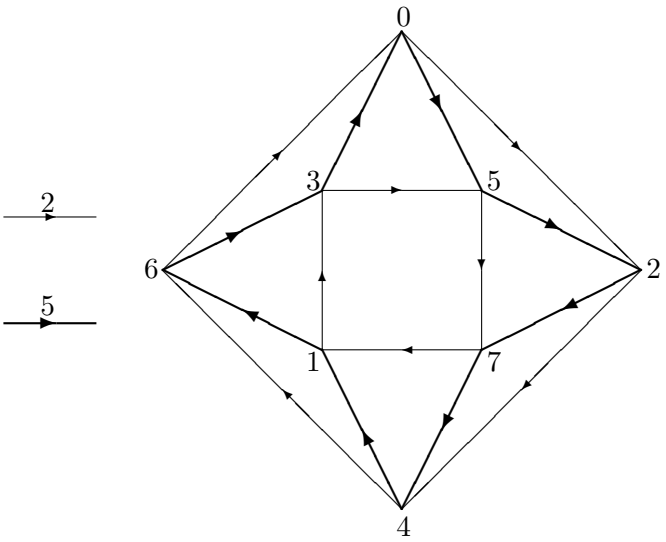
12. It is not commutative, for a followed by b leads to ab, while b followed by a leads to a^3b.

13. If more than one element of the cyclic group is used to generate the Cayley digraph, it may not be obvious from the digraph that the group is cyclic. See, for example, Figure 7.9, where 5 actually generates the group $\mathbb{Z}_6$ having these digraphs generated by 2 and 3.

14. No, it does not contain the identity 0.

15. (See the answer in the text.)

16. Here is a Cayley digraph.



17. **a.** Starting from the vertex representing the identity, every path though the graph that terminates at that same vertex represents a product of generators or their inverses that is equal to the identity and thus gives a relation.

b. $a^4 = e, b^2 = e, (ab)^2 = e$.

18. The diagram in Figure 7.13a which represents the Klein 4-group, and a square with solid clockwise arrows edges which represents $\mathbb{Z}_4$.

19. Generalizing Figure 7.13b, form a regular $2n$-gon with alternately solid and dashed edges, without arrows. The four properties listed after Example 7.10 in the text are satisfied and the digraph represent a nonabelian group of order $2n$ for $n \geq 3$.